

2023

PHYSICS — HONOURS — PRACTICAL

Paper : SEC-1P

Full Marks : 100

The figures in the margin indicate full marks.

SET - 3

[LNB - 20, VIVA - 20, PROGRAMMING - 60]

Answer *any one* question from *each group*.

Group - A

1. Plot the following functions for
- $-1 \leq x \leq 1$
- with same
- x
-
- y
- axes.

$$f(x) = \sinh(0.5x) \cos 2x$$

$$g(x) = 0.2x^2$$

$$h(x) = 0.1 \tan x.$$

Put the x -label and y -label as ' x ' and ' $f(x)$ ' respectively. Set the title of the plot 'function plots'. Save the plot in a suitable format. 20

2. Write the following data table in a file. Fit the following data-file with the function

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
y	0.0	0.214	0.216	0.154	0.091	0.045	0.018	0.005	-0.0	-0.002	-0.002

$f(x) = \sin(ax)e^{bx}$ for the parameters a , b . Write the values of a , b . Plot the data file along with $f(x)$. Save the plot in a suitable format. 20

3. Make the following data table in a file. Write a gnuplot script to fit the data with the curve

x	0.0	0.349	0.698	1.047	1.396	1.745	2.094	2.443	2.792	3.142
y	0.0	0.241	0.32	0.304	0.244	0.172	0.107	0.056	0.021	0.0

$f(x) = a \sin(x)e^{bx}$ having parameters a and b . Write the values of a and b . Plot the above data table along with $f(x)$. Save the plot in a suitable format. 20

4. Plot the following polar functions for $-\pi \leq \theta \leq \pi$.

$$\cos 2\theta, \sin 2\theta$$

Set the plot title 'Example of polar plot'. Set sample 1000. Save the plot in a suitable format. 20

5. Plot the following functions :

$$\begin{aligned} f(x) &= \frac{3\pi}{2} \frac{\sin x}{x} + \cos x \\ g(x) &= -1 \\ h(x) &= 1 \end{aligned}$$

with same x-y axes for $-2\pi \leq x \leq 2\pi$ with different line types. Set sample size 1000. Save the plot in a suitable format. 20

6. Plot the following piece-wise function using ternary operator.

$$\begin{aligned} f(x) &= -5, \quad x < 0 \\ &= 2x - 3, \quad 0 \leq x < 2 \\ &= x^2, \quad x \geq 2 \end{aligned}$$

Put the x-label and y-label as 'x' and 'f(x)' respectively. Set sample 1000. Save the plot in a suitable format. 20

7. Plot the following parametric functions using GNUPLOT :

$x(t) = 8 \sin t (2\cos^3 t - \cos t)$, $y(t) = 4 \cos 2t$ for $-\pi \leq t \leq \pi$. Set x-label as "t". Set title of the graph as "Parametric Plot". 20

Group - B

8. (a) Write a python program to determine the root of the following equation using Newton-Raphson method correct up to three decimal places of accuracy :

$$x^2 - 8x + 11 = 0$$

Take $x = 6$ as guess value. Print the root.

- (b) Define a python function for the following mathematical function :

$$f(x) = x^2 e^{-x^2}.$$

Find the values of $f(x)$ at $x = 1.2$ and $x = 2.1$ and print them.

30+10

9. (a) Write a python program to solve the following differential equation using Euler method :

$$\frac{dy}{dx} = xy, \quad 0 \leq x \leq 2$$

with initial condition $y(x=0) = 1$.

Calculate and print the value of y at $x = 2$. Take the increment of x as $dx = 0.001$.

- (b) Write a python program to find the factorial of the given number $n = 7$. Print the output. 30+10

10. (a) Using bisection method determine the root lying between $[3, 4]$ of the following equation up to three decimal places of accuracy :

$$2x^2 + 5 = e^x.$$

Print the root.

- (b) Write a python program to find the transpose of the following matrix :

$$A = \begin{pmatrix} 2.1 & 3 & 0 \\ 4 & 3.2 & 1 \\ -1 & 0 & 4.3 \end{pmatrix}.$$

Print the new matrix.

30+10

11. (a) Using bisection method find the root lying between $[1, 2]$ of the following equation up to three decimal places of accuracy :

$$\sin 3x + 3e^{-2x} \sin x - 3e^{-x} \sin 2x - e^{-3x} = 0.$$

Print the root.

- (b) Write a python program to find the following sum :

$$S = \sum_{n=1}^6 \frac{1}{n^2}$$

Print the output.

30+10

12. (a) Write a python program using the Newton-Raphson method to find the root of the equation

$$x^3 + (x + 1)^2 + 4x = 20$$

correct up to third decimal places. The root to be found lies inside the range $-2 < x < 3$. Print the output.

- (b) Write a python program to find the trace of the following matrix :

$$A = \begin{pmatrix} 1.7 & 2.8 & 3.9 \\ 4.2 & -5.3 & 2.1 \\ 7.1 & 8.2 & 9.3 \end{pmatrix}$$

Print the value.

30+10

13. (a) Using bisection method determine the root lying between $[1.3, 2]$ of the following equation correct up to three decimal places :

$$e^{3x} - 27x^6 + 27x^4e^x - 9x^2e^{2x} = 0.$$

Print the root.

- (b) Write a python program to calculate the sum of the following G.P. series :

$$S = 4 + 12 + 36 + \dots \text{ up to 5 terms.}$$

Print the output.

30+10

14. (a) Write a python program to take a positive real number x as input from the user. Define a new

variable $z = \frac{x-1}{x+1}$. Calculate the sum of the following series with an accuracy of 0.001.

$$s = \frac{1}{2} \left(z + \frac{z^3}{3} + \frac{z^5}{5} + \frac{z^7}{7} + \dots \right).$$

Print the sum and number of terms where the series is terminated with the aforesaid accuracy. Also print the value of e^s .

- (b) Consider the following two matrices :

$$A = \begin{pmatrix} 1.1 & 2.2 \\ 3.3 & 4.4 \end{pmatrix}, \quad B = \begin{pmatrix} -2 & 0 \\ 3 & 1 \end{pmatrix}$$

Write a python program to find $A - B$. Print the resultant matrix.

30+10